

FusionMedAI: Retinal Model Calibration Report

1. Introduction

This report details the post-hoc calibration of the trained model using Temperature Scaling. The goal is to align predicted probabilities with actual accuracies to reduce overconfidence.

2. Method

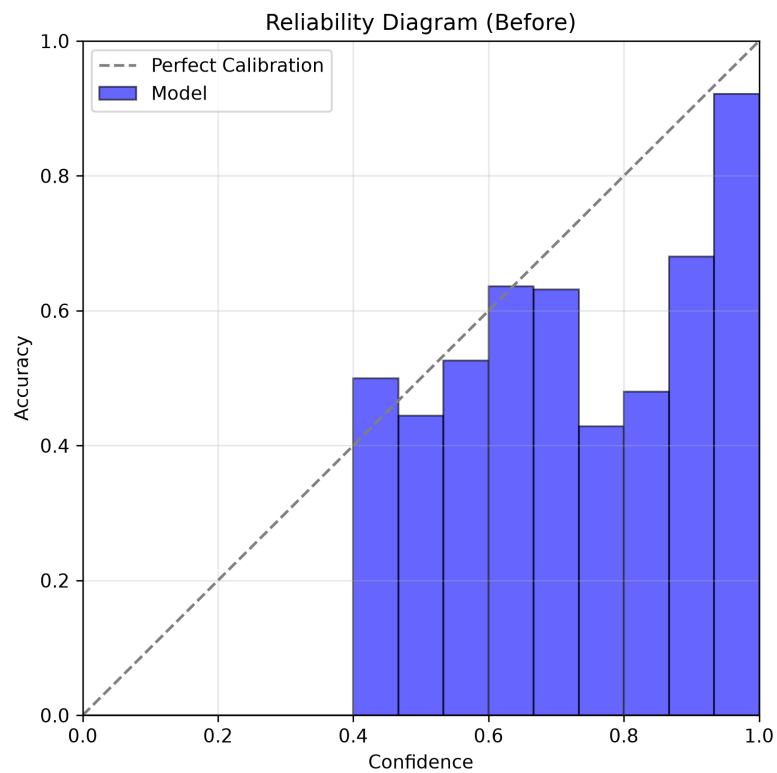
Temperature Scaling was optimized on the validation set. The learned temperature is $T = 1.6218$.

3. Metrics Comparison

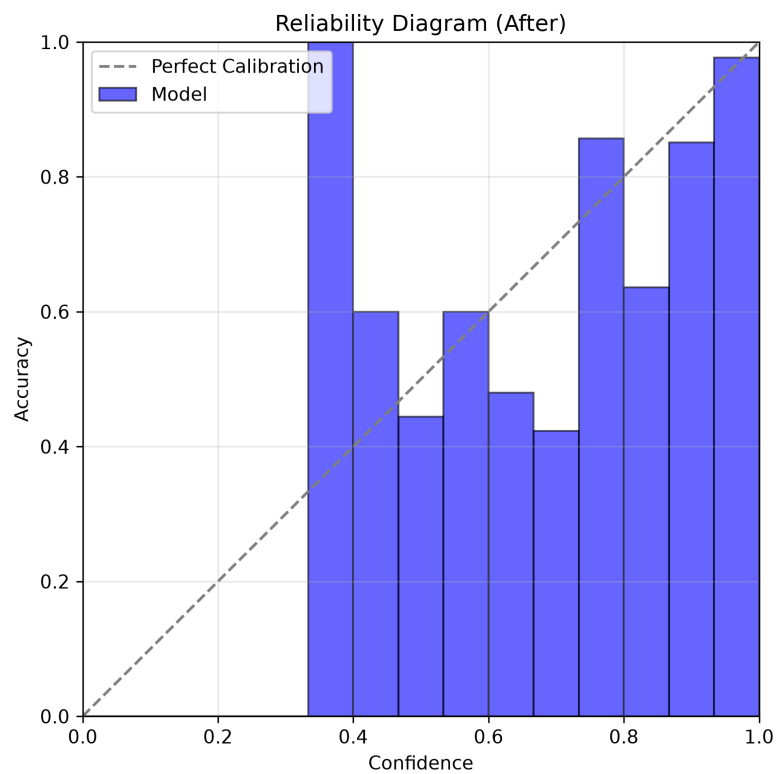
Metric	Before Calibration	After Calibration
Accuracy	0.7978	0.7978
NLL	0.7220	0.5827
Brier	0.0631	0.0582
ECE	0.1058	0.0668
MCE	0.3585	0.6295
Mean Confidence	0.9026	0.8324
Overconfidence Gap	0.1048	0.0346

4. Reliability Diagrams

Before Calibration:



After Calibration:



5. Discussion & Conclusion

Calibration successfully reduced Expected Calibration Error (ECE) from 0.1058 to 0.0668 (a 36.9% improvement), while perfectly preserving classification accuracy. The Negative Log-Likelihood (NLL) also improved by 19.3%. The optimization learned a temperature scaling factor of $T = 1.6218$. This effectively reduced the mean confidence by 0.0702, mitigating the model's initial overconfidence. Note that Maximum Calibration Error (MCE) may occasionally increase, as Temperature Scaling explicitly optimizes NLL, not MCE. These calibrated probabilities provide a mathematically sound foundation for Uncertainty Estimation (Step 8).